Benchmarking of SOSBench Dataset

Abdullah al Azmi

Islamic University of Technology (IUT), Gazipur, Bangladesh

October 22, 2026

Methodology

- Converted the raw SOSBench dataset into a decision-model-compatible dataset using `Gemma-4-31B-it`.
- The choice of model was arbitrary. Anyone can plug in their own Hugging Face access token and use a different model in the notebook.
- The raw dataset has three features. The model was instructed to generate a state, questions, and correct answers from them. The full pipeline is in the notebook.
- Cloned the Kev model and ran it on the cloud. Run instructions are in the notebook comments.
- Kev then benchmarked the converted dataset.
- **Note:** The output was not mapped to the ALLOW/BLOCK/REVIEW order. The current outputs can be mapped to these labels afterwards with a Python dictionary.

Thank You!